

1. DevOps.

2. GCP

3. Ansible

4. Docker.

5. Kubernetes (K8)

6. Terraform.

7. Grafana.

8. Dynatrace

* Release Management & Governance :-

① Controlled Rollout & Rollback strategies:-

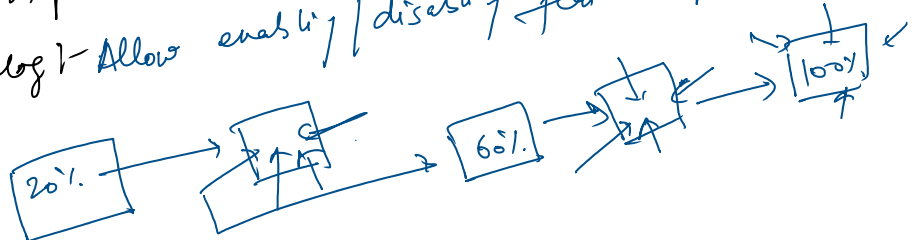
↓
ensure that s/w
updates or new features are
released gradually, manageable
manners.

Key Concepts:-

① Phased Rollout :- Deploying changes to a small % of users initially & expanding as per the stability.

② Feature Toggle / Flag :- Allow enabling/disabling features w/o code changes.

③ Canary Releases :-



④ Blue - Green Deployments :-

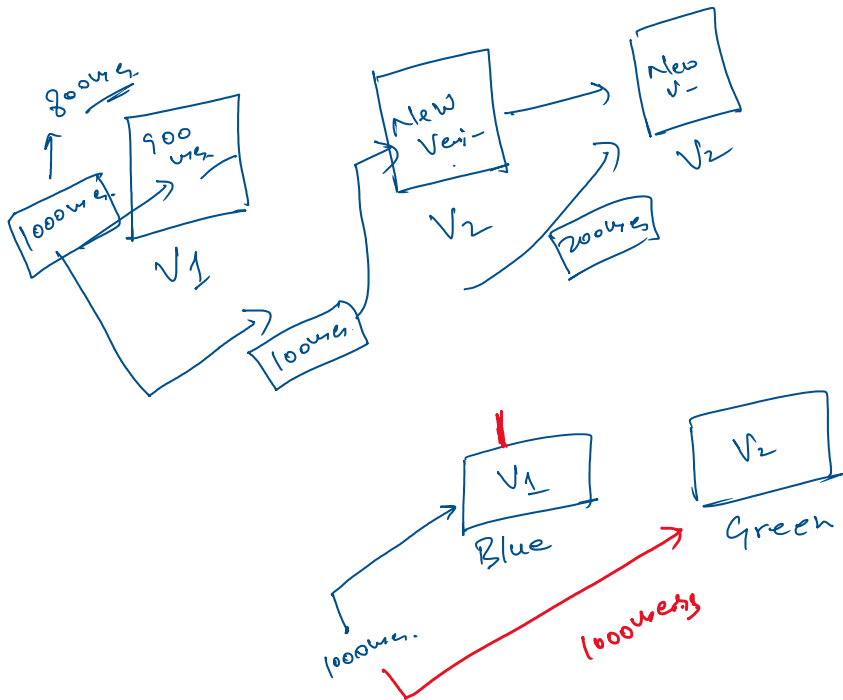
⑤ Rollback strategies:-

Manual Rollback -

Automated Rollback -

Re-deploy previous stable version if an issue is identified
Triggered when monitoring tool detects anomalies.
Maintain backward compatibility in schema for safe rollbacks.

Automated Rollback - 100%
 Database Versioning - Maintain backup change for safe rollbacks.



② Compliance in Release Process:-

- ① Documentation & Approval:- Each release should have review & approval from relevant stakeholders. documentation change ref.
- ② Audit Trails:- Maintain logs of all release activities for traceability & Audit
- ③ Change Management Policies:- Align ITIL
- ④ Regulatory Adherence:- HIPAA, GDPR or SOX.
- ⑤ Automation:- CI/CD Pipeline with built-in compliance checks ensure consistent & repeatable processes.

③ Business Impact Assessment:- Risk decision Making,

- ① Stakeholder Involvement - Business Owner, Product Manager & support team.
- ② Impact Analysis - Identify critical system or process that may be affected. Estimate downtime, if any & its effect on customer or revenue.
- ③ Communication Mapping - Understand upstream & downstream that could be impacted. Internal & external communication plan for

- (2) Imp-
- (3) Dependency mapping → Understand upstream &
- (4) Customer communication → internal & external communication expected change.

④ Risk Mitigation Plan:-

- ① Risk Identification → Use historical data, testing outcomes & stakeholders facilitate to identify risk.
- ② Risk Assessment → Evaluate likelihood & severity using risk metrics.
- ③ Contingency Planning → Define the Action will be taken if a risk materialize.
- ④ Mitigation Strategy → Pre-release testing, implement recovery & backup sys. before roll forward / roll back mechanism.
- ⑤ Monitoring & Alerting → observability tool Detect the issue early & act promptly.

GKE:-

- ① VPC
- ② Subnet.
- ③ router.
- ④ router-nat

- ⑤ VPC host shared.
- ⑥ iam binding &
- ⑦ service account.
- ⑧ Google container cluster.
- ⑨ Node pool
- ⑩ Public load balancer.
- ⑪ firewall.
- ⑫ firewall.tf.
- ⑬ Proxier.tf.
- ⑭ local.tf.
- ⑮ project.tf
- ⑯ vpc.tf
- ⑰ router.tf.
- ⑱ nat.tf.
- ⑲ shared-vpc.tf
- ⑳ Kubernetes.tf.